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On battery materials and methods

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ABSTRACT

Economical and efficient energy storage in general, and battery technology, in particular, are as imperative as humanity transitions to a renewable energy economy. Rare and/or expensive battery materials are unsuitable for widespread practical application, and an alternative has to be found for the currently prevalent lithium-ion battery technology. In this review article, we discuss the current state-of-the-art of battery materials from a perspective that focuses on the renewable energy market pull. We provide an overview of the most common materials classes and a guideline for practitioners and researchers for the choice of sustainable and promising future materials. In addition, we also discussed the best practice for battery performance testing and reporting.
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1. Renewables story

In 1959, global CO_2 levels were at 313 parts per million (ppm). Now, just six decades later, they are 100 ppm higher, recently surpassing 412 ppm in September of 2019 [1]. This is an unprecedented change in atmospheric conditions, which effect is already having, and will increasingly have a major impact on the Earth in the decades to come [2]. Despite advice from climate researchers to reduce this dangerously rapid rise in greenhouse gas emissions because of the uncertainty in the upper "safe" boundary [3], humanity continues to release roughly 40 billion tons of CO_2 into the atmosphere every year [4]. This is primarily an economically driven phenomenon, as fossil fuels have proven to be a relatively cheap and reliable form of energy. However, change is afoot. According to Bloomberg New Energy Finance's 2019 New Energy Outlook, renewable energy technology like solar and wind are already undercutting the cost of fossil fuels in two-thirds of all locations, and by 2030, it will be cheaper to generate from renewable sources almost everywhere [5]. We need this transition to happen as rapidly as possible if we are to prevent positive feedback effects and prevent the worst effects of climate change.

However, there are times when the sun doesn't shine, and the wind doesn't blow. This is why energy storage is needed, in order to provide renewable energy during these low production or high demand times, and also limit the wasted energy from renewable generation curtailment, which is becoming increasingly prevalent as renewable generation increases. Pumped hydropower has provided the cheapest form of energy storage in the past, with more recent projects ranging from $3.35 M to as low as $0.39 M per MWh for the addition of a pumping system to existing dams, the mean of these being about $1.8 M/MWh [6]. Pumped storage also has the added benefit of being more suitable for long term and general bulk energy storage. However, pumped hydropower has geographical limitations, namely a difference in elevation, which is not available in all locations. Electrochemical storage on the other hand is much more suitable for maintaining an uninterrupted power supply and providing load shifting. These are both important requirements with an increasing renewable generation [7]. These factors, along with Li-ion batteries plunging 85% in cost since 2010, to $187/kWh in March 2019, is why electrochemical storage has gained in popularity in recent years [8].

A recent study by Stock et al. [9] that looked specifically at the Australian energy landscape found that the country did not need significant amounts of new energy storage until roughly 50% renewable energy generation is reached. However, beyond 50% renewable energy generation, the amount of storage required increases significantly. Sisternes et al. identified that the "diminishing

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marginal value of energy storage means that the economically optimal penetration level of storage will be limited unless costs continue to decline well beyond current targets" [10]. This scenario is likely similar to other nations. As such, we need versatile and economically desirable forms of energy storage in order to bridge the latter half of the climb toward a 100% renewable economy and combat against the worst effects of climate change. We have until roughly 50% renewable generation to find and scale these solutions. The clock is indeed ticking.

2. A first principles approach

The energy storage market can be broadly split into two sectors: stationary and mobile. The stationary storage sector is purely interested in a cost per unit of energy ($/kWh) figure. Of course, safety is important too, but this is captured within the $/kWh figure of merit, as more dangerous solutions require more rigorous safety systems, which increase the cost. This is part of the reason why, in principle, nuclear power is cheap, but in practice, it is expensive—due to the high cost of safe construction [11]. Mobile storage, on the other hand, needs to factor in the energy density. It is impractical to have a one-ton battery powering a three-ton vehicle. Therefore a cost per energy density, $\frac{\$}{kWh/kg}$, figure is appropriate. Mobile storage is an area perhaps more interested in inherently safe systems, due to the impact of the weight of additional safety structures, as well as the additional cost.

2.1. Tools for low-cost batteries

Battery engineers have two broad strategies to achieve low-cost cells. Materials and morphology. Low cost, abundant materials that can be economically engineered into the appropriate form are required for low-cost cells. Therefore, any fabrication process that is itself inherently expensive, despite using abundant materials, must be excluded when engineering low-cost cells. However, materials that currently have a high production cost but are made out of abundant materials of the Earth, such as carbon nanotubes (CNTs), may start to become economically enticing as production economies of scale are reached in the coming years. Although currently expensive, methods of producing and sorting various types of CNTs are attracting huge research attention [12,13], and hence, prices on the order of $100/kg are anticipated in the future for multi-walled CNTs [14]. Fig. 1 shows the materials that are available for the production of a low-cost cell.

Currently, the battery industry is taking strides to move away from cobalt, due to its cost and ethical issues associated with its production [15]. Therefore, elements already more expensive than cobalt were excluded outright. The materials that are comparable in rarity or are currently cheaper but also rarer than cobalt are lightly crossed out with green. These elements may play some part in the batteries in a renewable economy, but only the noncrossed ones show promise for a truly supply unconstrained battery. However, chromium and vanadium may not be entirely supply unconstrained, as they are only roughly three to four times more abundant than cobalt [16,17]. Finally, the marginally toxic elements like vanadium, chromium, and fluorine are lightly crossed out with red, due to the health concerns when dealing with these elements. These environmental consideration of these elements needs to be on a case by case basis; V_2O_5, the most common oxide of vanadium, is environmentally toxic, while VO_2 is merely an irritant. In some countries, chromium use has particular health regulations in place due to the presence of trace amounts of Cr(VI), which is a carcinogen. While it would be ideal to avoid the toxic compounds altogether, if they showed enough promise, then they would still be commercially viable options. Fluorine demonstrates this perfectly. Compounds with fluoride in it are very often toxic; however, due to fluorine's abundance and usefulness in electrolytes due to its stability against oxidation, it has and will likely continue to play a major role in batteries.

When choosing a metal for use in a battery, there are a number of considerations to take into account. Lithium only comprises roughly 0.002% of the Earth's crust [16,17]. Any of the remaining groups of one or two metals underneath lithium in this adapted periodic table are tens to thousands of times more abundant while providing nearly as negative reduction potential, all between −2.37 and −2.93 V vs. the standard hydrogen electrode (SHE) compared to lithium's −3.04 V. This figure is important for achieving high voltages in cells; the more negative the standard reduction potential of the metal, the higher the voltage that the cell can theoretically have. Lithium will likely dominate the market over the next 5–10 years or so, but as demand for storage soars and lithium supplies dwindle, we will increasingly need to look at alternatives. Leisegang et al. [18] tried to quantify the electrochemical and economic benefits of each battery type, with the less well-known aluminum-ion coming out on top, due to its high theoretical volumetric energy density and being the most abundant metal in the Earth's crust.

3. Battery components

Generally speaking, a battery consists of five major components. An anode, cathode, the current collectors these may sit on, electrolyte and separator, as shown in Fig. 2.

The purpose of the anode is to hold the active ions in a high energy state. The higher the energy state, the higher the eventual voltage of the cell. In principle, pure metal is the best anode material, due to the metal being the highest possible energy state to hold the metal in, as well as holding the largest amount of metal atoms in the smallest amount of space. However, dendrite formation often hampers the use of pure metal. Hence, other solutions, which can prevent this dendrite formation, are often used instead, such as inserting the ions into the interlayer spacings in graphite. Conversely, when the metal is oxidized and the resultant ion moves over to the cathode during discharge, the ion inserted cathode needs to be in a much lower energy state than when it was at the anode. It is this difference in energy that gives the thermodynamic push for electrons to move through a circuit and do work. The larger the difference, the higher the voltage, and the battery will store more energy.

The electrolyte soaked separator must help facilitate this ionic movement while being electrically insulating in order to prevent short-circuiting. They should also be spectators on the electrochemical stage. If they do participate, they will be rapidly decomposed and/or coat either electrode in a solid electrolyte interphase (SEI), which can impede ionic movement, thus degrading the cell. The electrolyte can be engineered with additives in order to form stable ionically conductive SEIs [19–21].

4. Mechanisms

An ion-battery operates by transferring charge using freely moving ions shuttling back and forth between the two electrodes. This charge transfer occurs via different mechanisms depending on how the active ion interacts with an electrode material. The amount of charge that a battery can store depends on the potential difference of the electrochemical reactions of the active ion that occur at the two electrodes and the number of electrons involved in each of these reactions. One can either increase the potential difference between the two electrode processes or use ions involving multiple electrons to increase the amount of charge stored or the capacity of

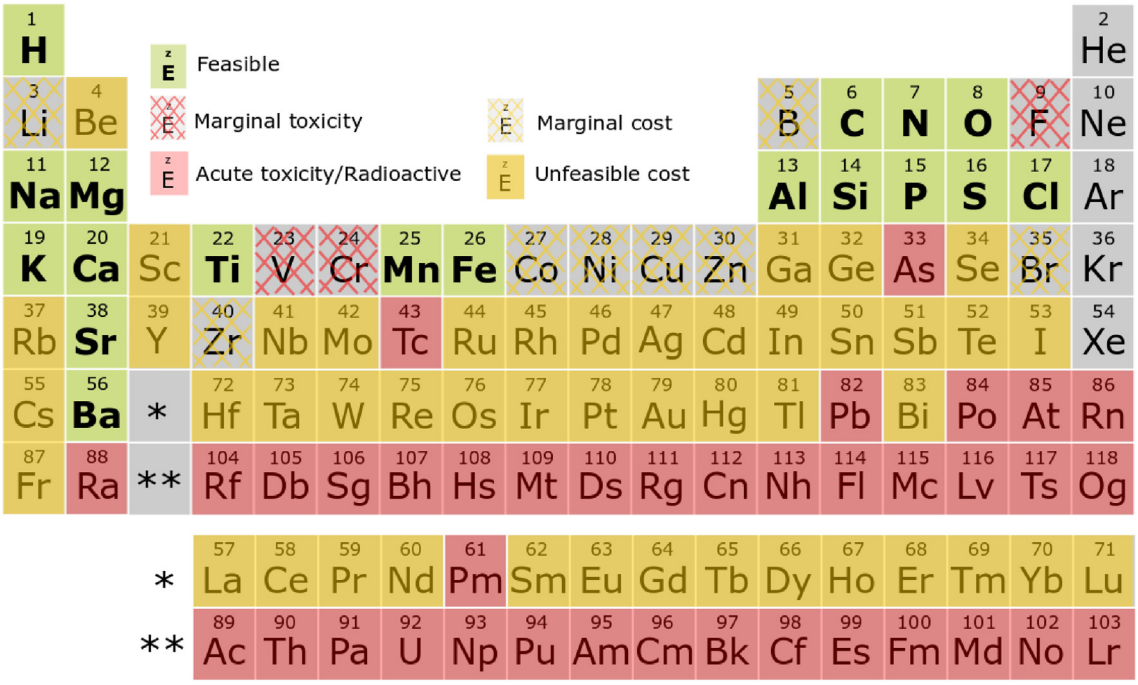


Fig. 1. The periodic table for cheap, safe batteries.

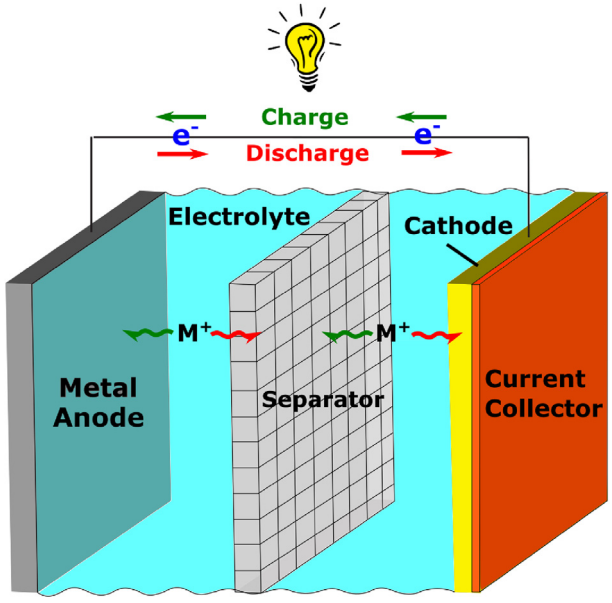


Fig. 2. A typical cell format. Charging processes are indicated in green, and discharging processes are indicated in red. On discharge, the high potential metal atoms oxidize, and the resultant ions move toward and interact with the reducing cathode.

the system. Both these factors are governed by the mechanism with which charge is transferred. The following sections discuss the common mechanisms of charge transfer in ion batteries.

4.1. Electroplating-stripping

The highly negative reduction potentials of alkali and alkaline earth metals have always been electrochemical processes of choice to capitalize on for high voltage systems. The low reduction potential for the Li^+/Li system at −3.04 V vs. the standard hydrogen electrode (SHE) has been one of the primary motivations behind the development of Li-ion batteries. In general, for any active ion M^{n+}, plating and stripping at the metallic M electrode generates maximum theoretical capacity, 3860 mAh/g in the case of Li^+/Li. However, with great reduction potentials come great reactivity. Operating with a highly reactive Li metal electrode is a risky business. Even if the sensitivity to O_2 and moisture is controlled, the reactive metal surface suffers from the formation of dendritic Li, which, in turn, leads to the shorting of the cells [22]. These aspects have been discussed for post-Li-ion systems in detail in Section 6.1. Despite significant benefits, plating and stripping of M^{n+}/M systems have often been replaced with intercalation-deintercalation systems, for example, graphite intercalation compounds (GICs). Although the theoretical capacity for GICs is much lower compared to the metal anode, the slightly more positive potentials of such compounds help avoid dendrite formation in most cases [23].

4.2. Intercalation-deintercalation

Intercalation of ions into layered materials, such as graphite or transition metal chalcogenides, has been extensively used as a charge transfer process in ion batteries [24–26]. Insertion/extraction of ions in the interlayer spaces of such layered structures usually results in no structural change in the host structure other than disruption in the interlayer spacing (usually an increase in the van der Waals gap) [27]. In such a case, the ion is said to have intercalated into or deintercalated from the layered structure. The energy change involved with intercalation and thereby the disruption of the layer spacings is mediated by charge transfer between the host ion and the guest structure, usually a redox process. In the case of layered materials, such as graphite, redox reaction upon intercalation involves the π-electrons of the material [28]. In the case of transition metal chalcogenides, the redox reaction involves the transition metal center. These two types of intercalation-deintercalation processes are well exemplified by the $LiCoO_2$-graphite system introduced by Goodenough and coworkers in 1980 [29].

During charging, Li^+ ions are removed from the $LiCoO_2$ cathode and intercalated into the graphite anode. This movement of Li^+ ions results in the oxidation of Co(III) centers in the cathode to Co(IV) and the reduction of graphite at the anode. The oxidation of Co(III) to Co(IV) is a thermodynamically unfavorable process, and the large potential difference of it against the reduction of graphite upon Li^+ ion insertion results in the large cell voltage. Thus, the high oxidation potential of a transition metal center is exploited by using intercalation/deintercalation of ions to generate high cell voltages. Although intercalation mechanisms have proven to be a reliable way of storing charge with a decent capacity and good cyclability, it is rather inefficient in terms of atom economy. Lithium deintercalation in $LiCoO_2$ is reversible only to up to $Li_{0.5}CoO_2$ [30]. Therefore, for every unit charge stored reversibly, we need two Co atoms and four O atoms. This results in lower specific capacities of intercalation electrodes, especially for electrodes, such as $LiCoO_2$, which are limited by the growing prices of Co.

Moreover, intercalation/deintercalation of larger guest ions, such as Na^+ and K^+, or multivalent ions, such as Mg^{2+} or Al^{3+} is not as straightforward as that for Li^+ [31]. For example, Na^+ intercalation into graphite is negligible [25,32] and multivalent ions have poor mobility in most layered frameworks due to their higher charge density.

These shortcomings of intercalation/deintercalation mechanisms have led researchers to look for other processes to achieve higher capacities. To obtain higher capacities, one has to involve more electrons per atom in the redox processes. However, such processes usually change the chemistry and structure of electrode materials largely, thereby affecting reversibility.

4.3. Conversion

A conversion mechanism involves electrochemical reactions that are leading to the formation of new chemical species, often with structures different from the initial structure. For example, a conversion reaction for a Li-ion system can be represented as follows:

$$M_aX_b + (b \cdot c)Li^+ + (b \cdot c)e^- \rightleftharpoons aM^0 + bLi_cX$$

where M is a transition metal (TM), X is an anionic species, for example, oxide, fluoride, or sulfide. Unlike in the case of intercalation mechanisms, the TM center is reduced to zero oxidation state, thereby involving more redox electrons, and thus, a larger theoretical capacity [33].

The first report for a conversion-based Li-ion battery was from Tarascon and co-workers in 2000, where they reported a series of transition metal oxide nanoparticle-cathodes with capacities of up to 700 mAh/g over 100 reversible cycles [34]. The system was based on the formation and decomposition of Li_2O from Li^+ ions aided by the reduction and oxidation of the TM center, respectively. Following are the reactions:

$$\text{Cathode}: \ CoO + 2Li^+ + 2e^- \underset{\text{charging}}{\overset{\text{discharging}}{\rightleftharpoons}} Co^0 + Li_2O$$

$$\text{Anode}: 2Li \underset{\text{charging}}{\overset{\text{discharging}}{\rightleftharpoons}} 2Li^+ + 2e^-$$

By returning to the concept of atom economy, for the same transition metal, the conversion mechanism draws two unit charges per cobalt atom as opposed to one per two Co atoms in the intercalation/de-intercalation mechanism. However, the higher capacities come at the cost of transforming the electrode material, which results in numerous drawbacks, such as low conductivity of the resultant electrode species, volume expansion leading to poor cycling stability and high-voltage hysteresis resulting in low efficiency [24]. Conversion reactions based on the oxidation/reduction of the anionic species do not require the TM and can be generalized by the following reaction;

$$X + aLi^+ + ae^- \rightleftharpoons Li_aX$$

The chalcogens (O, S, Se, Te) and the heavier halogens (Br, I) undergo conversions of this type to form Li_2X and LiX, respectively [33]. Such conversions have high theoretical capacities, for example, 1672 mAh/g for Li—S systems. However, the poor conductivity of elemental sulfur and the related sulfides, and dissolution of the polysulfide species affect efficiency and cyclability drastically [35].

4.4. Alloying

Another mechanism of charge transfer, *alloying*, can be seen as a variation of the conversion mechanism, the difference being, instead of forming a chemical species Li_aY, a lithium alloy Li—Y is formed. Alloying materials, preferably having low de-lithiation potentials, are mostly used in anodes. Si, Sn, and Sb are the most extensively studied alloying anodes for Li-ion batteries [23]. For example, a Si anode can alloy Li with the stoichiometry of $Li_{4.4}Si$, leading to a capacity of around 4200 mAh/g. However, alloying reactions suffer from a similar flaw to conversion reactions, a drastic transformation of the electrode material. In this case, the electrode material undergoes a large volumetric expansion. Lithiation to the extent of $Li_{4.4}Si$ ($Li_{22}Si_5$) is accompanied by volume expansion of 300% [36]. Silicon electrodes are further discussed in 6.1.6.

Volume changes of such severe magnitudes often crack the electrode material, leading to loss of contacts, loss of active material, disruption of the solid electrolyte interphase (SEI) resulting in electrolyte decomposition (see Section 6.2.1). Even if the lithiation is controlled to $Li_{3.75}Si$ ($Li_{15}Si_4$) the volume expansion is still significant (290%) [37,38]. Alloying materials are often made into composite structures to accommodate expansion and increase conductivity, for example, as composites of porous carbons, graphene, CNTs, etc. However, the addition of high surface area carbonaceous materials often leads to a pseudocapacitative behavior. Although strategic nanostructuring can help to avoid pseudocapacitance, these additives lead to a loss in specific capacity compared to the pristine material and also add to the cost.

4.5. Pseudocapacitance and surface mechanisms

In order to resolve the expansion issues in alloying electrodes, many groups have resorted to nanostructuring of electrode materials. However, high surface area electrode materials or composites are not always ideal battery materials. High surface area materials tend to exhibit pseudocapacitative behavior. Pseudocapacitance can be seen as an intermediate case between bulk redox in crystalline materials and surface-induced capacitance in layered materials. Pseudocapacitance is a faradaic process, however it involves redox reactions at surface sites, which are usually disordered. In common battery materials, redox reactions occur uniformly across a crystalline phase, and thereby, at the same potential. On the other hand, as surface redox sites are nonuniform, pseudocapacitative charge transfers occur within a range of potentials, thereby exhibiting a sloped charged-discharge characteristic or broad, ill-defined peaks in cyclic voltammetry [23]. Such materials increase the rate capability as the surface involvement reduces the solid-state diffusion lengths or even have diffusionless transformations. Notable here is the report by Grimaud, wherein the surface of a

MnO–LiF composite undergoes a diffusionless phase transformation through which Li^+ ions can be reversibly exchanged [39]. Such surface-based conversions can have the benefit of tuning the capacity by controlling the particle size or morphology, and thereby, the surface availability. However, such processes are still conversions and will come with the associated disadvantages. One specific drawback is the electrolyte decomposition that occurs when the solvent is oxidized at the surface at high potentials. In general, while dealing with new nanostructured electrode materials, it is advisable to consider pseudocapacitance and surface mechanisms especially, because this classification is important from an application perspective.

4.6. Conventional, dual-ion and mixed-ion batteries

While traditional ion batteries are based on cationic active species, progress in research has led to the development of systems based on or involving anionic active ions. More unconventional systems, such as the dual-ion or mixed ion batteries, have also become increasingly popular. The working principles of such systems have been compared to conventional cation-based *rocking-chair* type batteries. Also, the chloroaluminate-based battery that involves an unusual electrolyte deposition reaction has been discussed.

4.6.1. Aluminum ion/Chloroaluminate batteries

Aluminum, being a trivalent metal, can exchange three electrons, thereby having a high theoretical capacity of 2980 mAh/g. However, oftentimes Al^{3+} is not the active species in these batteries. The high charge density of the trivalent Al^{3+} (54 pm ionic radius) compared to monovalent Li^+ (76 pm ionic radius) results in a high propensity of the ion to interact with the host lattices and electrolytes. This makes the diffusion of the trivalent ion complicated and often counter anions, such as Cl^- form complexes with Al^{3+} to form $AlCl_4^-$ which is also an intercalating ion. For example, in the aluminum-graphite system, the following reactions occur [40];

$$\text{Cathode}: C_nAlCl_4 + e^- \underset{\text{charging}}{\overset{\text{discharging}}{\rightleftharpoons}} C_n + AlCl_4^-$$

$$\text{Anode}: Al + 7AlCl_4^- \underset{\text{charging}}{\overset{\text{discharging}}{\rightleftharpoons}} 4Al_2Cl_7^- + 3\,e^-$$

It is to be noted here that, unlike in conventional cation-based batteries, the active ion being an anion results in graphite being the cathodic species. The intercalation of $AlCl_4^-$ at graphite means that only one e^- can be exchanged, which nullifies the primary advantage of using trivalent aluminum species as active ions. However, reports of reversible Al^{3+} intercalation in cathode materials, such as V_2O_5, TiO_2, and more recently, potassium-rich cryptomelane have been made, although competitive metrics have not been achieved [41–45] (see Fig. 3).

4.6.2. Mixed-ion batteries

The inefficacy of Na^+ ion intercalation in common host materials, as well as the low degree of Na^+ ion storage in most materials, have prohibited the popularity of Na^+ ion systems. However, in 2013, Liu et al. came up with the concept of using more than one active cation to circumvent the Na^+ ion problem. They reported a Li^+/Na^+ mixed-ion electrolyte, which utilized a Na^+ ion insertion/extraction cathode and a Li^+ ion extraction/insertion anode [46]. The working principle of this battery, as depicted in Fig. 4, differs from the conventional *rocking-chair* mechanism, as two different cationic species are simultaneously active on opposing electrodes.

The total capacity of an ion battery is influenced by both the capacity of the cathode and the anode. Mixed ion batteries can get around the problem of having to find a high capacity anode and cathode for the same ion and allow researchers to cherry-pick high-performance electrodes individually.

Since the initial report, several similar systems based on other cations, such as K^+ and Mg^{2+} have been reported [47]. Such systems can be seen as alternatives to Li^+ ion batteries for which we are closing in on resource limitations.

4.6.3. Dual-ion batteries

Although the intercalation of anions into graphite has been known since the 1930s [28], only since the 90s anions were used as active species in batteries. Carlin et al. [48] developed dual-graphite batteries with various molten salts of imidazolium cations and anions, such as PF_6^-, BF_4^-, $AlCl_4^-$, $CF_3SO_3^-$, and $C_6H_5CO_2^-$. Since then, such batteries that employ both anions and cations of an electrolyte as active species have been developed, which have now become popular as dual-ion batteries. Considering the vast research of Li intercalation into graphite, most of these systems use Li^+ as the cation, while PF_6^-, BF_4^-, $TFSI^-$, F^-, ClO_4^- have been the popular anions [49]. In a dual ion battery, during charging, the cations intercalate into the anode while the anions intercalate into the cathode, simultaneously. During discharge, the ions deintercalate into the electrolyte. The charge separation created by these simultaneous intercalations can be seen as the mode of storage for such systems. Such systems usually have higher operating voltages compared to the respective cation only systems, usually above 4.5 V [49]. As battery research has made progress over the years, dual-ion batteries have started employing charge transfer mechanisms beyond graphite intercalation to alloying reactions as well. To move away from Lithium, researchers have employed organic cations with PF_6^- anions for dual-ion systems as well [50].

5. Best practice: reporting battery metrics

Although it seems obvious that battery metrics have to be comparable between different laboratories and battery types, there is a surprising diversity of methods for obtaining these metrics. For example, the specific capacity of an electrode is typically reported in mAh/g. While current and time are unambiguous, the mass of the electrode has been published as a mass of active material, mass including additives (binder, conducting carbon), or other variations. In this section, we will discuss key battery metrics and will suggest meaningful methods for reporting them.

Charge/discharge plots (curves) contain the key metrics for batteries. Fig. 5 shows the first three charges/discharge cycles of an aluminum-ion battery using a MoO_3 cathode at a rate of 40 mA/g. This example is typical, and we will use it to demonstrate the use of different battery metrics.

5.1. Coulombic efficiency

The Coulombic efficiency, or charge/discharge efficiency, is the ratio of the charge extracted from the battery (C_d) over the charge injected to the battery (C_c) over one cycle:

$$\eta = \frac{C_d}{C_c} \times 100 \tag{1}$$

Charge/discharge curves, as depicted in Fig. 5, record the time multiplied by the (usually constant) charging/discharging current $i_{c/d}$ [mA/g] on the x-axis. Given that current is defined as flow of electric charge per time, and the fact that the electrode mass does not change (constant mass), the Coulombic efficiency η of a charge/

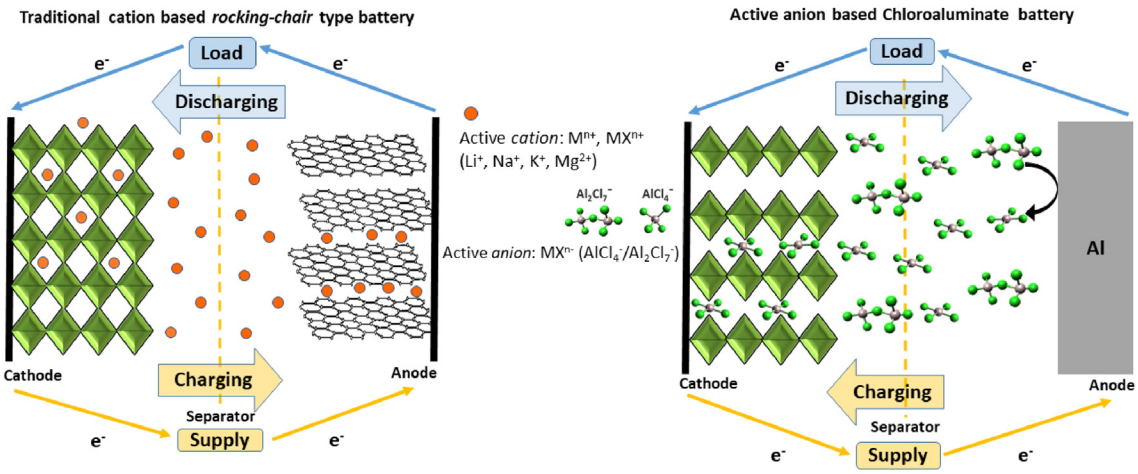


Fig. 3. Working principles of conventional cationic rocking chair batteries and the chloroaluminate battery.

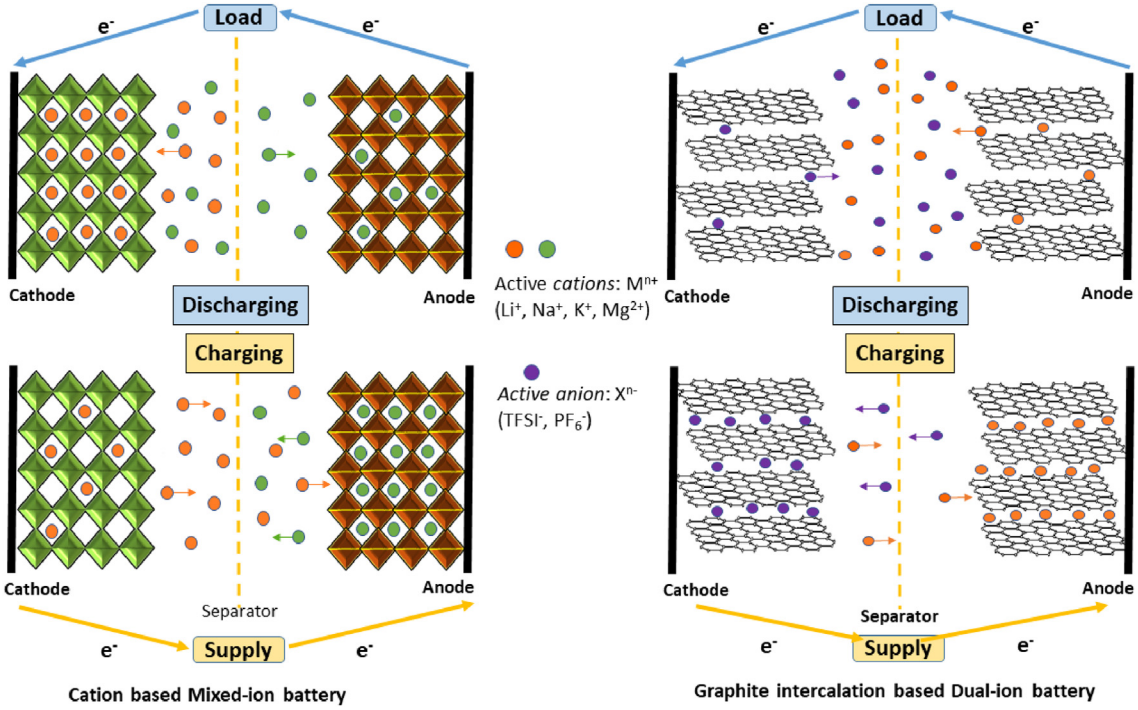


Fig. 4. Working principles of mixed-ion and dual-ion batteries. **Left:** Mixed ion battery mechanism where one cationic species comes out of an electrode and a different cationic species inserts in the counter electrode. **Right:** Dual ion battery mechanism where the cations and anions are pulled apart to opposite electrodes on charging, then fall back together into the electrolyte on discharge.

discharge cycle can be simply calculated by dividing the specific capacity at the end of the discharging time by the one at the end of the charging time. In the example above, C_c of the first cycle is 88.64 mAh/g and C_d was found at 78.65 mAh/g. Thus, the Coulombic efficiency of this cycle can be calculated to $\eta = 88.7\%$.

It is worth noting that the software of most battery testers offers this option automatically. Also, since the Coulombic efficiency can be directly measured, not requiring any assumptions by the researcher, it is a straightforward and unambiguous metric.

Coulombic efficiency is a measure of how much charge can be recovered from a battery relative to how much was required to charge it. In other words, it quantifies the amount of charge that drives irreversible side reactions in a battery. Typically, η should be very close to 100%. The Coulombic efficiency of the second and third cycles in the example of Fig. 5 can be calculated to 94.4%, 95.6%, respectively. This is a very typical example, where some energy is required in the first cycle to equilibrate the battery chemistry. Then, η approaches 100% in the following cycles.

5.2. Capacity

In section 5.1, it has been discussed that the Coulombic efficiency of a battery charging/discharging cycle can be calculated unambiguously from the specific capacities at the end of each charge/discharge curve. However, the specific capacity, and other derived metrics depend on the mass in the denominator. Unfortunately, there are no clear guidelines on how to determine the correct mass for this value.

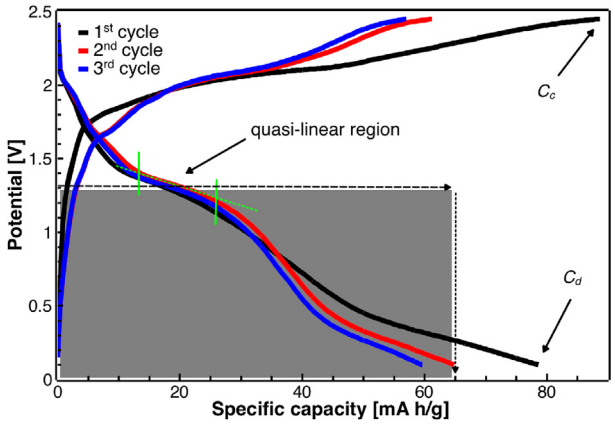


Fig. 5. Typical battery charge/discharge curves. The example shows the first three cycles of an aluminum-ion battery using a MoO_3-based cathode and a charge/discharge current of $i_{c/d} = 40$ mA/g.

From a user perspective, it would be useful to report the specific capacity relative to the mass of the whole device. This is what vendors and end-users should do. However, since the masses of battery casings, current collectors, and other battery components can vary substantially without affecting the performance of the battery, this would not be meaningful for research and development purposes.

On the other hand, typical electrode materials contain an active material (for example graphite), as well as additives, such as conducting carbon and binders. Reporting the mass of the active material would only be equally meaningless, since additives cannot be omitted in an actual device. Therefore, we recommend to base the specific capacity on the combined masses of anode or cathode materials, including additives.

The mass of the electrolyte should also be omitted in research reports because the volume of electrolyte required depends on the fabrication and configuration of the device.

5.3. Voltage

When discharging a battery, the voltage constantly changes and in a nonlinear fashion (cf. Fig. 5). However, the voltage is an important parameter that goes into derived metrics as outlined in section 5.5. Unfortunately, there are different methods in the literature on how to determine the voltage, for example, by using the voltage where the charging and discharging curves intersect [51]. It can be easily seen in Fig. 5 that this method overestimates the voltage significantly—it would be at about 1.7 V in this example.

In a typical battery, after an initial nonlinear potential drop, the discharge voltage enters a region where it decreases linearly. At the end of this *quasi-linear region*, the voltage drops further, rapidly. In practical applications, only the quasi-linear region is useful. Fig. 5 shows an example of a quasi-linear region. Following this logic, the average of the quasi-linear region is a good estimate for the battery voltage. In the above example, this voltage can be found at approximately 1.3 V. More accurately, the voltages at the beginning and the end of the quasi-linear region can be provided.

5.4. Cycle life

The cycle life represents the number of times a battery can be charged and discharged over its lifetime. According to the industry standard, a battery has reached the end of its lifetime, when the (specific) capacity has reached 80% of its "initial" value. Since batteries require one to five cycles in order to equilibrate the battery chemistry, the "initial" capacity should be recorded after these equilibration cycles.

It has to be noted that the cycle life of a battery is heavily dependent on the depth of discharge (the percentage of the battery's total capacity that is being used in one cycle). It is, therefore, necessary to report not only the cycle life of a battery but the cycle life at the respective depth of discharge. Realistically, a battery is only used within its quasi-linear potential range. For the example in Fig. 5 this means that a realistic depth of discharge would be roughly at 40% (discharge until 25 mAh/g at a total capacity of 65 mAh/g for the second cycle). The capacity at the end of the quasi-linear region is called *maximum reversible capacity*.

5.5. Energy density

The energy density of a battery describes how much energy the device can store per mass or volume. Thus, the energy density can be simply calculated by multiplying the specific capacity by the voltage. If we use the second cycle of Fig. 5 as an example, the energy density of this battery would compute to $1.3\text{V} \times 65\text{mAh/g} = 84.5$ mWh/g, indicated by the grey rectangle in the figure. Although this is the common way to calculate the energy density, it has to be noted that this is a very optimistic approximation, given that the correct energy density equals the area under the discharge curve. Furthermore, as discussed above (sections 5.2 and 5.3), the electrode mass and battery voltage can be determined in different ways, which makes it imperative to report how these values have been measured, if the energy density is being reported.

For end-users, the useable capacity (or energy density) is much more relevant. This can be calculated by multiplying the specific capacity (energy density) by the depth of discharge or multiplying the maximum reversible capacity by the voltage. In the example used here, the useable energy density is thus 33.8 Wh/kg.

5.6. Factoring in cost

Although this is not relevant for purely scientific work, the cost of storing energy is of huge significance for end-users. A useful metric to factor in the cost of the materials is the *cost of useable capacity/energy*. In scientific work, it is important to be able to directly compare the performance of the active materials themselves, as various research groups may use different additives. Therefore, the mass and cost units have to be based on the active materials of both cathode and anode, as well as the minimum amount of electrolyte required. The cost of useable capacity/energy can then be determined by dividing the cost of materials per unit mass by the useable capacity/energy. Including the performance and cost characteristics based on the total electrode or cell is also an honest way to accurately assess the current economic viability of the product.

5.7. Power density

The power density of a battery is the maximum power that a battery can deliver based on its mass (W/kg in SI units). The power density depends on the kinetics of the charge migration and the transfer of the device. For example, batteries, where faradaic processes are involved, generally have a lower power density compared with supercapacitors, where charges are just transported. Similarly, systems involving bulky ions or highly viscous electrolytes have lower power density than those featuring highly mobile charge carriers. Power densities can be measured by cycling the battery at different charging/discharging rates and then working out the maximum power density of these experiments.

6. Promising electrode materials

6.1. Anodes

The use of metallic anodes in ion batteries has been the subject of much research for a very good reason. In order to maximize both specific and volumetric capacities, one must be able to pack the atoms into as little space as possible. The most efficient way to pack metal atoms is as the pure metal solid, therefore, using the pure solid as an electrode material will give the highest theoretical capacity available.

The reduction potential of the $M^{n+} \rightarrow M$ reaction is also usually very high e.g. $Li^+/Li = -3.04$ V, $Na^+/Na = -2.71$ V. This not only means that a lot of charges be stored per gram or per cm^3, but the charges can also be stored at a high energy, leading to higher energy densities overall.

Unfortunately, the use of metallic anodes is fraught with difficulty. While each metal has its own quirks, the problems mostly stem from the surface of the metal being very reactive. The advantages and disadvantages of some of the most common anodes are summarized below.

6.1.1. Graphite

Graphite is perhaps one of the most successful and attractive battery materials found to date. Not only is it a highly abundant material, but it also helps to avoid dendrite formation and the high reactivity of alkali metal anodes. Not to mention the fact that it is naturally conductive is also a huge positive. Another advantage is that it is very easy to study material with many techniques, such as Raman spectroscopy, X-ray diffraction, XPS, and more being utilized with relative ease.

Graphite is not only the benchmark nonmetallic anode for lithium systems, but it is also the benchmark cathode for conventional aluminum ion batteries. The maximum stoichiometry achieved for lithium is LiC_6 giving a theoretical capacity of 372 mAh/g. For aluminum ion batteries, it is the $AlCl_4^-$ ion that intercalates [52]. However, the specific capacity in this system is much lower, usually between 60 and 70 mAh/g [18,25,40].

The general mechanism is also fairly simple. Ions intercalate into the layers of carbon atoms, which expands the lattice. As the ions deintercalate, the lattice relaxes back, thus making way for a highly reversible process [25,53]. Fig. 6 shows two different models for intercalation. The Rüdorff model has ions intercalate fully along layers one at a time, whereas the Daumas-Hérold method shows partial intercalation in different layers in an ordered fashion before both finish at the fully metalated “stage 1” complex.

The formation of these graphite intercalation compounds (GICs) is a charge transfer process where electrons are transferred into the graphite molecular orbitals giving the structure a negative charge. This can then interact with the positively charged ion [28]. Interestingly, in aluminum ion batteries, graphite is used as a cathode, and so the same phenomena occur with active species of the opposite charge. Instead of electrons being transferred into graphite, giving it a negative charge, the electrons are stripped away, giving it a positive charge. This allows for the negative $AlCl_4^-$ ions to intercalate in the same way as conventional positive ions do when graphite is used as the anode [52].

Interestingly, graphite has not found such widespread success in either sodium or magnesium ion batteries despite the similarities in size and or charge to lithium. In the efforts to explain this, a quantum mechanical study was carried out by Liu et al. [32]. The results of this study showed that Na and Mg generally have the weakest binding for a given substrate arising from a competition between the ionization potential of the metal ion and the ion-substrate coupling.

The intercalation of sodium into graphite can, however, be achieved through intercalation of solvated ions [54–56]. By chelating the cations with linear polyethers, such as diglyme (bis(2-methoxyethyl) ether), stable complexes resistant to reduction can be formed. While successful, this method greatly reduces the theoretical capacity of the cell. Instead of intercalating a single sodium ion, a large complex must now intercalate, which will take up much more space and have much more mass.

While not technically graphite, hard carbons have also received extensive attention over the years. These materials are usually produced by pyrolyzing cellulose at high temperatures (> 1000 °C) [57]. These gained most attention after their successful implementation in sodium-ion batteries by Dahn and coworkers [58]. This work showed successful intercalation of sodium ions at a capacity on the order of 300 mAh/g. However, this capacity faded quickly, limiting its commercial viability.

However, not all graphite is created equal, and different types of graphite have shown to have different levels of performance. For example, Wang et al. [59], investigated many different forms of graphite as a cathode in aluminum ion batteries. Kish graphite flakes (a highly crystalline form of graphite made as a byproduct of the steel industry), natural graphite flakes, highly oriented pyrolytic graphite (HOPG), spherical graphite powder (potato-shaped graphite), as well as acetylene black, were all compared. See Fig. 7.

Efforts to improve upon graphite have been made; however, they have been met with limited success. For example, “graphite foam” is a material that has been investigated, both as a free-standing electrode material [60], as well as a support onto which materials may be coated [61,62]. Graphite foam is produced by expanding the interlayer spacing of graphite, allowing for an increased surface area while maintaining high conductivity throughout. The increased gap between the carbon layers can also facilitate easier intercalation into the structure, as demonstrated by Wu et al. [60] and Lin et al. [52].

Functionalizing graphite can also have an effect on performance. In particular, graphite fluorides have been extensively studied for lithium-ion and aluminum ion batteries [63–66]. The process of fluorination can increase the surface area of the graphite leading to an increase in observed capacity [65]. Surface fluorination can also have an effect on the formation of the solid electrolyte interphase [64].

Battery research has not been immune to the “graphene boom” in recent years [67–72]. However, it is hard to imagine these systems making it to market any time soon.

Overall, graphite represents a benchmark to which all other materials should be held to. Not only is it effective in terms of the underlying electrochemistry, but it is also incredibly cheap and easy to process. Being naturally conductive does not hurt either. This means that the active material does not need to be “diluted” with an extra conducting additive, such as “Ketjen Black” or “Super P” conductive carbon when electrodes are prepared.

6.1.2. Lithium metal

Lithium metal, in particular, has caused many problems over the years. The idea of using lithium metal in a cell was proposed and demonstrated by Whittingham in the 1960s [73].

One of the most difficult aspects of using lithium metal to navigate is the formation of dendrites on the surface upon plating and stripping. Dendrite formation is a complex process, and many theoretical models have been developed to try and describe the phenomenon and hopefully suppress it [74–77].

In short, in standard electrolytes, lithium is deposited onto the metallic surface in an asymmetrical fashion. Instead of plating evenly, the fresh lithium deposits as fractal spikes on the surface. If

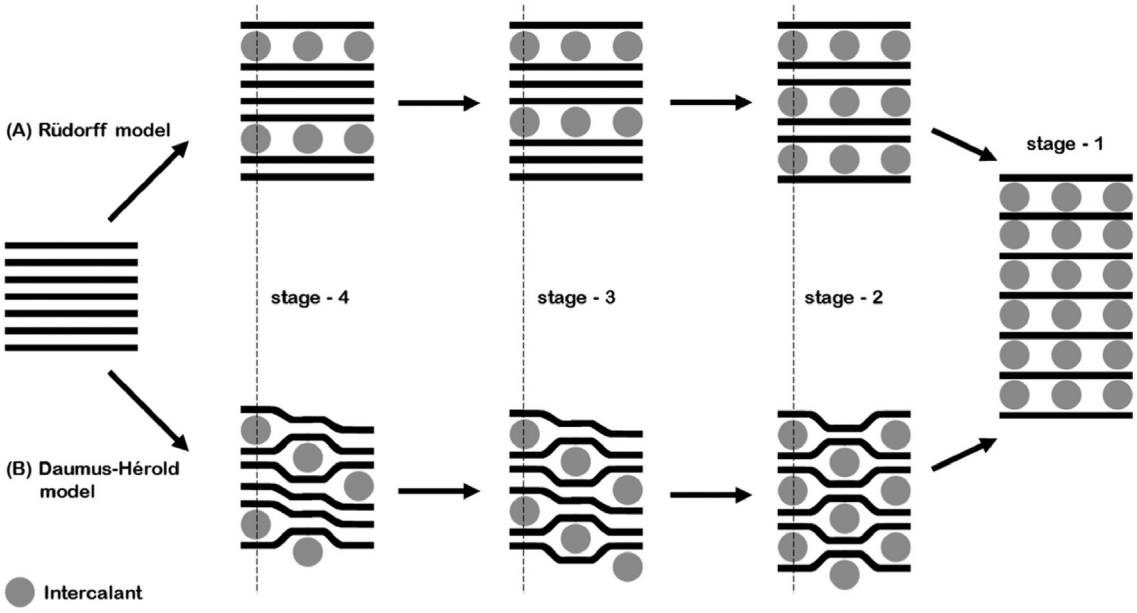


Fig. 6. Two opposing mechanisms for the formation of graphite intercalation compounds. Upon removal of the ions, the scheme progresses in the backward direction. Reproduced with permission from Ref. [53] Copyright 2014, Royal Society of Chemistry.

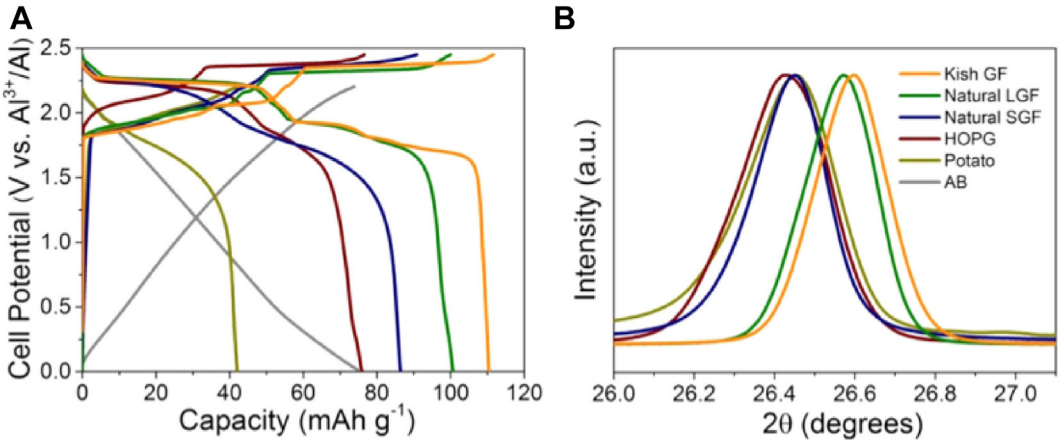


Fig. 7. A: Charge discharge curves for various types of graphite used in the study. Voltage plateaus all appear at roughly the same potential, but the capacity is vastly different. B: X-ray diffraction studies of the different forms of graphite, this peak is related to the inter-layer spacing the structure. The colors in the legend represent identical materials throughout both plots. Adapted with permission from Ref. [59] copyright 2017 American Chemical Society. Further permissions related to the material excerpted should be directed to the ACS.

these metallic dendrites grow large enough, they can puncture the separator and short the device [22,78].

As such, dendrite suppression is an active research area with many different techniques being discovered; for example, coating the anode with various materials, such as carbon nanospheres [79], 2-dimensional materials [80,81], or polymers [82] to form an artificial SEI. Alternatively, electrolyte additives, such as other lithium salts or organic carbonates, have also been shown to suppress dendrite growth [19,21].

While unlocking the true capability of lithium metal is a worthy goal. The low relative abundance makes it unattractive in the long term. In the short and medium term, however, lithium will likely continue to dominate the market. As a mature battery technology that has excellent metrics, it will no doubt remain a benchmark for many years to come.

6.1.3. Sodium metal

Given that sodium sits below lithium in the periodic table, it is substantially more reactive than lithium, which makes it difficult to work with. A fully developed battery using metallic sodium does exist in the form of Na/S batteries.

The Na/S system traditionally uses a solid beta-alumina electrolyte and operates at a temperature of between 300 and 350 °C [83]. This is required for the movement of the active species through this solid electrolyte, but also has the side-effect of keeping the sodium anode in the molten state (Na m.p. = 98 °C). This, of course, presents considerable safety concerns.

To date, little progress has been made in terms of a room temperature secondary battery using sodium metal. Work has been done on the plating and stripping of Na metal in organic electrolytes, but molten salt systems are the main system used [84–87].

Cao et al. [88] have reported a functioning room temperature sodium secondary battery by using 4 M sodium bis(fluorosulfonyl) imide in a variety of ether solvents. The sodium is initially plated onto a copper current collector and plated and stripped in subsequent cycles.

The main aspect of working in sodium's favor is its natural abundance compared to lithium. In all other aspects, sodium is left wanting. Its redox potential, while close, is not as low as lithium, and its theoretical capacities are much lower owing to sodium's much larger mass and size.

6.1.4. Magnesium metal

One of the often stated advantages of magnesium metal batteries is that magnesium metal is "dendrite free," unlike its alkali metal counterparts [89–91]. This is often attributed to the higher rate of self-diffusion of the divalent metal on the surface. However, recent work has shown this to be untrue (at least under certain circumstances) [92,93].

In particular, Davidson et al. have shown, via calculations, that if the rate of the electrochemical reactions (dictated by the charging rate) is sufficiently high, one can expect dendritic growth to occur [92]. Fig. 8 shows the results of these calculations. Regardless, magnesium metal still has a large theoretical capacity (3833 mAh/cm^3 and 2205 mAh/g) and so getting it to work in a cell would be an admirable goal.

In spite of its seemingly dendrite free nature, magnesium metal is probably one of the most difficult battery materials to work with. Like all of the metal surfaces, it is highly reactive, and most electrolytes spontaneously decompose on to form a "solid electrolyte interphase" or SEI [94]. This is also observed in Li and Na systems, but the crucial difference with Mg is that the SEI layer is ionically insulating [89,95]. This means that once the SEI layer is formed on the Mg surface, the battery is essentially dead as the charge cannot flow across the boundary. As a result, the majority of magnesium battery research has focused on electrolytes that do not spontaneously decompose to form this SEI layer [89,91,94–99].

While initial systems focused on Grignard reagents, the field has settled on ethereal solvents, primarily tetrahydrofuran (THF), but glymes have also been used more recently [99,100]. This SEI formation also rules out a lot of commonly used salts, such as PF_6^- or ClO_4^- as magnesium sources, which compounds the challenge further [95].

Work has gone into alternative anodes for magnesium ion batteries, but as always, the advantages of using the pure metal are hard to ignore.

6.1.5. Aluminum metal

Aluminum metal, while less prone to spontaneous combustion than lithium or sodium, is still a highly reactive metal surface. In the air, this is quickly passivated to form aluminum oxide, which prevents further corrosion and allows for easy handling and processing. Similar to magnesium, this passivation layer is ionically insulating. Likewise, much research has gone into the development of electrolytes that are able to break down the passivated surface to perform electrochemistry with the reactive metal underneath [101].

Current literature on Al-ion batteries are biased toward a eutectic mix of $AlCl_3$ and 1-ethyl-3-methylimidazolium chloride ([EMIm]Cl). These studies represent about 2/3s of all academic studies [18]. The chlorine from the [EMIm]Cl is more attracted

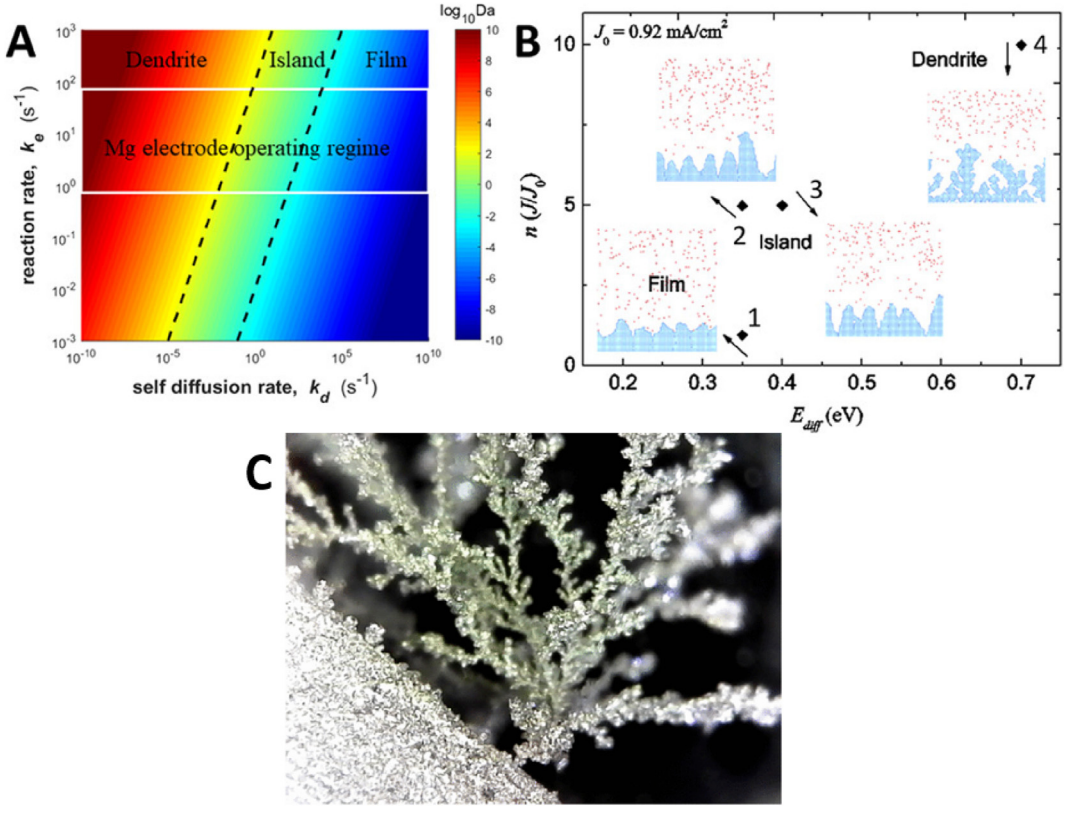


Fig. 8. A: Calculation comparing the rate of self-diffusion k_d and the rate of reaction k_e with the morphology of deposition. B: Kinetic Monte Carlo simulations for different current density and diffusion barriers. C: A digital photograph of magnesium dendrites obtained during these experiments. Adapted with permission from Ref. [92] copyright 2019 American Chemical Society.

to the Lewis acidic $AlCl_3$, so an ionic liquid comprised of $EMIm^+$ and $AlCl_4^-$ ions are formed. Once the ratio of $AlCl_3$ exceeds 1:1, $Al_2Cl_7^-$ species are produced, which render the ionic liquid Lewis acidic, and it is then able to penetrate the oxide layer to the fresh, reactive Al metal underneath.

However, this [EMIm]Cl-based electrolyte is expensive, which severely limits the commercial potential, despite often being coupled with a cheap graphite-based cathode. Also, current collectors for the cathode are often based on expensive molybdenum or tungsten metal, so an overhaul of the electrolyte and current collector are needed for commercial viability. Currently, Al-graphite batteries are limited to energy densities, similar to lead-acid [102]. So, if energy densities greater than lead-acid are desired, a cathode material with the ability to reversibly intercalate the highly charge dense Al^{3+} is needed.

6.1.6. Alloying electrodes

As stated, using the native metal as the anode material in a secondary battery is the benchmark anode in all cases. However, as outlined, the native metals often come with practical problems stemming from their extreme reactivity. Using an anode capable of forming an alloy with the metal, for example, Na_3Sn is another alternative that often leads to high capacity systems.

As mentioned in section 4.4, various metals have been investigated for this purpose, but the most common across all ion systems are Si, Sn, and Sb. The main issue for all of these metals is the large volume expansion sustained upon metalation. For example, going from Sn to $Li_{4.4}Sn$ shows a 260% volume change [103], going from Sn to $Na_{15}Sn_4$ shows a 420% volume change [104]. The larger the active ion, the greater the volume expansion.

This volume change causes the "pulverization" of the material, which impacts the electrical contacts and causes the capacity to fade rapidly [105–107]. This is shown in Fig. 9, where on the initial charge, the material swells and cracks. An SEI is also formed at this stage. Upon discharge, the material shrinks, cracking even more. This is exacerbated over many cycles.

Once the contact is broken, the ions become stuck in the lattice and cannot flow out again. One novel solution to this problem by Ryu et al. was to apply pressure to the cell, which pushed the particles closer together and restored electrical contact [108].

More delicate methods have also been applied to try and solve the volume expansion and its associated problems. The general approach involves the formation of some form of composite, whether it be intermetallic, for example, SnSb, Cu_2Sb [109–114], or carbon based [103,115–119].

Another aspect that is often explored along with these composites is nanostructuring. These solutions do show promising results; however, in terms of long-term application, caution must be taken. As previously mentioned, materials like carbon nanotubes should be removed from consideration in terms of scale and quality of production.

Out of these anodes, the only real long-term competitor appears to be silicon. Tin is often said to be a cheap material, but in terms of its crustal abundance, it sits well below lithium [16]. Silicon, on the other hand, is the second most common element in the crust below oxygen.

6.2. Cathodes

6.2.1. Sulfur

Sulfur is one of the holy grails for battery materials. It is not only abundant, but it also boasts of a maximum theoretical capacity of 1675 mAh/g (this assumes that each sulfur atom gains two electrons during discharge). Compared to the standard $LiCoO_2$, which has a theoretical capacity of 274 mAh/g, although only half the lithium can be removed in practice, meaning that the capacity is limited to half that figure [30]. The electrochemical potential, combined with sulfur currently being in oversupply [120], makes it a very attractive option.

However, sulfur is not without issues. The three main problems holding back sulfur are its low conductivity, the large volume expansion upon reduction, and the polysulfide "shuttling" phenomenon [30]. For amending the conductivity, the sulfur is often diluted with a conductive material like Ketjen Black, carbon nanotubes, or other carbon-based conductive matrices in which the sulfur can sit [30,121–123]. This is a common practice for many materials, but it is still an important consideration, especially in light of the volume expansion issue. Upon complete reduction to Li_2S, the volume of the sulfur cathode is expected to increase by more than 170% [30]. This places large mechanical strain on the cathode, which can result in cracks. This disrupts the connection between the conductive carbon and the sulfur, causing degradation of performance.

The final drawback of sulfur is the polysulfide shuttle effect. Sulfur as a material is made up of S_8 rings that undergo sequential reduction to S^{2-}. First, the ring is split, making S_8^{2-} with a stoichiometrically appropriate amount of the active ion. In the case of monovalent cations, one cation simply attaches at either end of the sulfide chain [30]. In the case of magnesium, on reduction a Mg^{2+} ion keeps the ring intact by the two end sulfides bonding with it to form MgS_8 [124]. Next, the sulfur is progressively reduced, and the chain shortens to S_x^{2-} (where $3 \leq x < 8$) are formed. It is these intermediate species, which are often soluble in the electrolyte [30,125]. The final step is further progressive reduction all the way to S^{2-}. Fig. 10 illustrates the progressive reduction pathway of the S_8 ring. The lower order polysulfides where $x < 3$ tend to be insoluble in most conventional electrolytes [30,125]. Exactly which polysulfides are soluble in the electrolyte will depend on the electrolyte used. How this progressive reduction mechanism works in the case of the trivalent cations, such as Al^{3+}, is not fully known and is not discussed in the literature as far as the authors can see. However, there will have to be more than one sulfide chain involved in order to satisfy the stoichiometry.

There are various methods to suppress polysulfide shuttling, which broadly fall under two strategies, namely, cathode engineering and electrolyte engineering. Cathode engineering generally

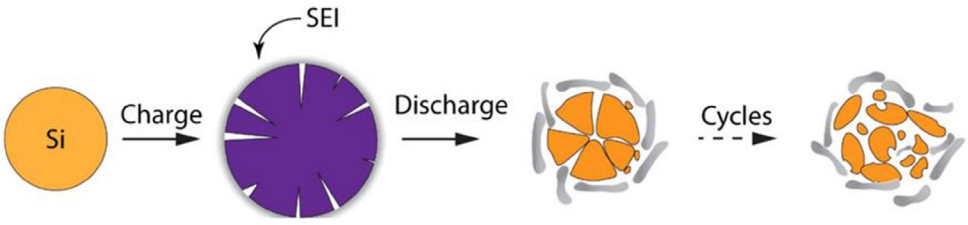


Fig. 9. Showing one of the main failure modes of electrode materials that undergo a large volume change. As the material is metalated, an SEI grows, and cracks appear in the structure. Upon demetalation, the material shrinks and cracks, which affects performance greatly. Adapted with permission from Ref. [107] copyright 2107 Springer Nature.

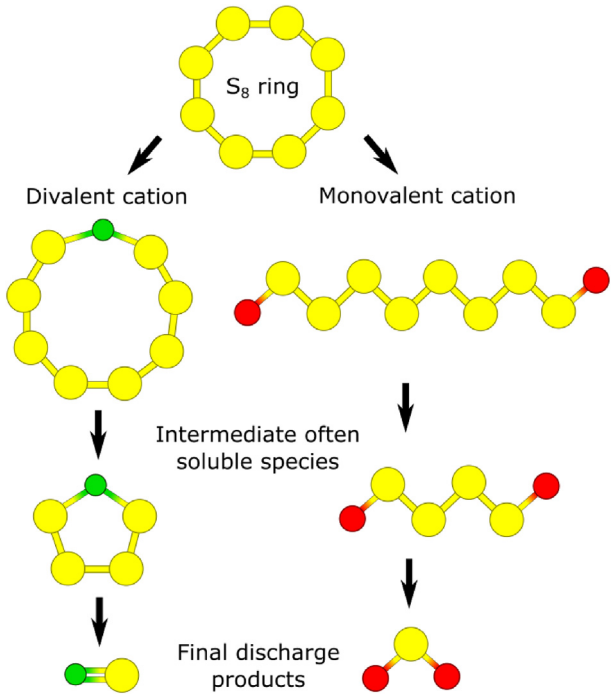


Fig. 10. Progressive reduction pathway of the S_8 ring to S^{2-} sulfide ion. Some intermediate steps omitted to keep the diagram succinct.

involves the trapping of the polysulfides by providing a physical barrier, which is still conductive to the active ions, or by anchoring the polysulfides. The anchoring occurs through attractive forces, such as Coulombic attraction, with the addition of functional groups. Some barriers include porous conductive carbon matrices [126,127], core-shell structures [128] or alucone coating [129].

Anchoring can be achieved through sulfur-polyacrylonitrile composite nanofibers [130], copper nanoparticles [131], or even metal-organic frameworks (MOFs) that have coordinately unsaturated metal sites (often abbreviated as CUMs) [132]. These CUMs can provide sticky Lewis acidic sites for the polysulfides to adhere to [133,134]. Alternatively, MOFs can be used as a precursor to highly porous, conductive, carbonaceous structures after pyrolysis [131,133].

Electrolyte engineering provides another pathway for the reduction of the shuttling phenomena. Using a solid, ionically conductive electrolyte means that the polysulfide species do not have a solvent to dissolve/disperse into. Other solutions involve the use of additives, such as bis(4-nitrophenyl) carbonate. This has been shown to precipitate out the polysulfide species in Li—S cells, largely eliminating the shuttling phenomena and allowing Li—S cells to retain over 700 mAh/g after 300 cycles [35].

6.2.2. Air cathodes

Cells are usually closed systems, where the active ions flow from one end to the other. Metal-air batteries flip this idea on its head using oxygen (ideally from ambient air) as the cathode material. The oxygen is either reduced to superoxide O_2^- and/or oxide O^{2-} upon discharge and then oxygen O_2 is regenerated upon charging. Oxygen is the single most abundant material that could be used as a cathode; hence, if it could be paired with an abundant metal anode in the earth, then it could turn out to be the cheapest cell chemistry of all.

The most studied batteries of this type is the Zinc-air and Li-air battery. Other metals have been used, such as Mg and Al, but these are only known as primary cells, and so are beyond the scope of this article. Like its group neighbor sulfur, the capacity of this anode is extremely high; however, there are still many factors impacting the performance of the cathode. In particular, the morphology of the carbonaceous air-electrolyte interphace has a huge influence on the end performance of the battery. As one might expect, the surface area of the substrate is proportional to the true capacity of the cell [136]. However, this is not the full story as for Li-air cells, pore size is a more important factor [135] smaller pores can be clogged more easily with the various lithium oxide precipitates that form during discharge [135]. Increased structural complexity usually comes hand in hand with increased cost, and so care must be taken to stick to cheap and effective methods of obtaining the desired morphology.

In the case of Zn-air batteries, an aqueous solution of 30% KOH is used as the electrolyte. This serves many purposes. First, it significantly reduces the rate of hydrogen evolution at the Zn metal surface [137]. Unfortunately, it is still not enough to make the reaction nonspontaneous under these operating conditions (Zn/ZnO $= -1.26$ V vs. SHE, $H_2O/H_2 = -0.83$ at pH 14). This concentration of KOH also shows the greatest electrolyte conductivity [138].

Perhaps the most promising, low cost, and environmentally benign metal-air battery is Fe-air and Fe—OH. Usually, metal-air batteries show poor cycling efficiency. This limits commercial viability due to energy being a relatively expensive component in the operating cost over the life of a battery. If energy is wasted, then this cost component increases. Generally, Fe-air-based batteries manage efficiencies of 55—70% [139,140]. However, Manohar et al. have shown that Fe—OH batteries can have efficiencies as high as 96% [139] with the use of carbonyl iron coated via electrodeposition with bismuth sulfide. Carbonyl iron is a highly pure form of metal, and this is a common anode in high-efficiency iron batteries [140]. Without Bi_2S_3, Manohar et al. achieved an efficiency of 89%, while after the electrodeposition of Bi_2S_3, efficiency as high as 96% was achieved. The high efficiency is due to Bi_2S_3 forming elemental Bi on charging. This increases the overpotential for hydrogen evolution, effectively suppressing the unwanted phenomena. So, although Fe-air batteries generally have a lower capacity than other metal-air batteries (Manohar et al. achieved 300 mAh/g with their cell), the extremely high abundance of iron, coupled with one of the highest metal-air Coulombic efficiencies make it a great candidate for further research and development. In fact, Manohar et al. estimated that at commercial volumes, their battery could reach costs as low as $3/kWh. This is a figure that is nearly two orders of magnitude below 2019 prices, which were about $187/kWh on average [8].

In general, metal-hydroxide batteries may be preferable to metal-air ones. Generally, metal-air batteries require the use of expensive catalysts, which negates the benefit of cheap electrode and electrolyte materials [140,141].

6.2.3. Metal chalcogenides

Most “standard” battery cathodes fall into this category. Ranging from V_2O_5 to $MoSe_2$ many compounds have been tested over the years in all the different types of batteries. This article will be limiting the discussion to elements identified in the first principles chapter, namely Ti, V, Mn, and Fe and their oxides and sulfides. In general, however, these materials possesses both a high redox potential, moderately high capacities, and fast, reversible reactions.

6.2.4. Titanium oxides

Titanium dioxide is mostly known as a white pigment for paint or as an active layer for dye-sensitized solar cells [142]. However, it has also found use as a battery material. Interestingly, titanium

oxides have been used primarily as anodes where most other transition metal chalcogenides act as cathodes.

As with most materials, the bulk of the research is based on lithium. The insertion voltage for TiO_2 is higher than that of graphite, ca. 1.5 V vs. Li^+/Li compared to 0.2 V vs Li^+/Li for graphite [143]. While this does lower the voltage of the battery, it can avoid the formation of SEI layers, which usually form at lower voltages [144,145].

The main advantage of the titanate spinels as anodes is the increased surface area compared to the traditional graphite. This means that the possible charging and discharging rate is much higher than traditional cells [146–148] As a result of the increased cycling capabilities, lithium titanate cells have found applications in some electric vehicles, where having a high power density (W/kg) is just as important as having a high energy density (Wh/kg).

The decreased voltage of lithium titanate cells does decrease the energy density of the cells. A pleasant side-effect is that it makes them safer [149,150]. In traditional Li-metal or Li-graphite cells, the lithium atoms are stored in the anode with a very high potential making them much more reactive with air or moisture. With a lower potential anode, cell failure can be less catastrophic.

The cycling performance of these cells is also improved compared to other systems with cells in literature achieving 3000–7000 cycles [151–153]. As lithium becomes more and more expensive, cell lifetimes are going to play an ever more important role.

Fast charging rates are becoming increasingly important in battery technology as the push toward electric vehicles continues. While clever engineering of a battery management system can improve charge times, the underlying chemistry is a major factor, and lithium titanate cells appear to have found their commercial niche in this space.

6.2.5. Vanadium oxides

Vanadium oxides as a material have been used in all of the major ion batteries at some point in their research history. It is generally a cheap material, with vanadium(V) oxide being the cheapest. The price generally increases with decreasing oxidation state, and as a result, this discussion will be mostly centered around V_2O_5.

On top of this, many attempts have been made to engineer a large variety of nanostructures for this material ranging from nanoflowers to nanorods to xerogels [154–162]. As with most materials, lithium has found the most success with V_2O_5 and due to its small size and low charge, it is able to intercalate up to two Li-ions per V_2O_5 unit reversibly [163,164]. A third Li can also be intercalated, but this is not a reversible process due to a phase change [163,164]. Orthorhombic V_2O_5 is the most common polymorph to use because of its structure.

With the higher charge density ions Mg^{2+} and Al^{3+}, there is little to no reversible insertion achieved [165]. This is due to strong interactions with the highly ionic lattice [157]. It is not purely a size effect as both of these ions have a smaller ionic radius than Li^+.

Recently there has been some controversy with the V_2O_5 in magnesium ion batteries. It has been observed that using hydrated V_2O_5 increases the observed capacity of the cathode material. It was reasoned that the greater ability of the hydrated lattice to "screen" the charge dense magnesium ions allowing them to flow more freely through the structure [155,166]. However, Lim et al. [165] showed through an extensive study that the reversible capacity stems mostly from proton insertion and while Mg^{2+} is inserted it is only to a very low degree $Mg_{0.17}H_xV_2O_5$ ($0.66 \leq x \leq 1.16$). This work also disproved the hypothesis of the cointercalation of species, such as $Mg(H_2O)_y^{2+}$. This research also highlights the need for high levels of proof for claims surrounding an intercalation mechanism. Electrochemical results are not enough to prove a mechanism, and at a bare minimum, X-ray diffraction studies must also be undertaken.

A similar result was observed for other transition metal oxides [34]. This means that in the future, as we move away from lithium-ion batteries, we may also have to move away from these materials. As previously discussed, some of these problems can be avoided with nanostructured materials; however, in the long term, it may be better to investigate other materials entirely.

Beyond V_2O_5, VO_2 has also gained attention recently as a potentially viable material [167]. The crystal structure of VO_2 is shear-type, and so is more tolerant of ion insertion and removal [168,169].

Because transition metals are not naturally conductive, they often need "diluting" with binders and conductive carbon when used as electrode materials. One way around this, as demonstrated by Dai et al. [170], where a composite material of VO_2 and reduced graphene oxide was made. This eliminated the need for a binder and also made the resulting zinc ion batteries flexible on top of allowing for fast charging. While impressive, the use of reduced graphene oxide presents issues of scalability.

6.2.6. Transition metal sulfides

One way to increase the mobility of charges within a lattice is to use softer anions that can polarize more easily and allow passage of intercalating species. Moving to transition metal sulfides accomplishes this goal. Selenides and tellurides would have an even greater effect, but these are excluded based on their abundance. While moving down a group does increase the polarizability of the anion, it also increases the atomic mass of the anion, which means that the theoretical capacity is generally lower for these materials compared to their oxide counterparts. However, theoretical capacity is only part of the story.

While ion mobility in the sulfides may be higher, the voltage of these electrodes is also lower [171]. This leads to trade-offs in both capacity and voltage, which greatly reduces the energy density of batteries that use sulfides as cathode materials.

The most commonly investigated transition metal sulfide is molybdenum disulfide MoS_2. With its layered structure, similar to that of graphite, it has found a lot of success across a range of ion batteries [172–175]. However, the low abundance of Mo makes it hard to consider for applications in the long term. Of the transition metals identified as abundant enough to be useable in the long term, titanium disulfide TiS_2 is the most studied material owing to it being the first lithium ion battery material. MnS_2 VS_2 and FeS_2 have also been used, but in a very limited fashion [70,176–180].

Titanium disulfide was the first cathode material used in rechargeable Li-ion batteries by Whittingham [73]. It was chosen for being both the lightest and the cheapest transition metal available. Its layered structure allows for the reversible intercalation of metal ions. However, it fell out of favor quickly due to its relatively low voltage and poor cycle life [176]. Due to it being a semi-metal, TiS_2 has a high conductivity which may make it attractive for solid state batteries [181].

Transition metal chalcoginides have been the prototypical cathodes for ion batteries for as long as ion batteries have existed. They have a crystalline structure, which can go through a phase transition to accommodate other ions and their redox reactions are easy to detect and understand. However, it is important to consider the abundance of potential materials available when building economically viable systems.

Caution must also be excised when choosing from the wide variety of morphologies of these materials available. Nano-architectures are an attractive solution to gain some extra

performance, although if the synthesis of complicated composites is too difficult or time consuming, then it may not be an economically viable solution.

6.3. Other cell components

While the most attention in battery research is paid to the active materials and the electrolytes, a fully commercialized battery has many more components than just those. Inside the cell, separators and current collectors play crucial, yet often under-appreciated, roles. The material that encases the cell must also be considered for cost and ease of use.

6.3.1. Separators

Separators serve as a barrier between the anode and cathode, which prevents a short circuit. However, they must still be permeable to allow the active species to flow between the electrodes. Since, in their most simple incarnation, they exist only as a spacer, it is most important that they are inert under operating conditions.

On the small/lab scale, glass microfiber is a common material, but for larger scale and commercial products, carbon based polymers, particularly polyolefins, are the standard. Commercial systems also use separators as a fail-safe mechanism. If the temperatures get too high, the polymer melts, which stops the flow of ions between the electrodes effectively shutting the battery down [182].

From a materials perspective, there is no current threat to the supply of separator components. However, when separators develop further to become thinner or allow for easier flow of the active species, it is important that this perspective is not lost. In particular, the environmental impact of functionalized polymers is a concern [183]. There may also be a drive to use materials from more eco-friendly sources, such as cellulose [184].

6.3.2. Current collectors

In a battery, the role of the current collector is to collect and deliver electrons from and to as close to the active material as possible. It is usually a metallic substrate on which the active material is attached. The main requirements for current collectors are that they are conductive and stable/inactive under the operating conditions inside the cell. Current collectors are at least partially exposed to the electrolyte, and so they must be carefully selected, or else they will corrode quickly [185].

In lithium ion systems, aluminum is commonly used as the cathodic current collector. This cheap metal is readily passivated by the PF_6^- ion, which prevents further corrosion [186]. At the anode, however, the favored material is copper foil, which is far more costly.

In standard aluminum ion systems, a corrosive ionic liquid electrolyte is used. This means that conventional materials, such as stainless steel, copper, and nickel, cannot be used as they are quickly corroded. On the lab scale, researchers have turned to much more expensive metals, such as molybdenum or tungsten, to solve this issue. An in-depth cost analysis by Munoz-torrero et al. showed that in cells using these current collectors, they made up 53% and 70% of the cost of the final cell, respectively [187].

In the ideal case, some variants of steel would be suitable; however, steel is often very susceptible to corrosion. Some research groups have attempted to use carbon-based current collectors; however, to get the required conductivity, they have had to resort to using carbon nanotubes [188]. While these can impart interesting properties, such as flexibility to the cell, as previously discussed, they are not suitable for large scale applications.

As with separators, development of current collector's area is relatively slow, but as shown above, this often overlooked component can be incredibly important in the quest toward the maximum energy density. The use of better current collectors is tightly tied to the development of new electrolytes that are compatible with cheaper metals.

6.3.3. Binders

When preparing an electrode for use in a cell, the active material is combined with some highly conductive carbon to increase the conductivity and a binder to improve mechanical properties and increase adhesion to the current collector.

The binder is another often overlooked component of a battery, but on larger scales, it too will need to undergo closer scrutiny. The most common binders used are polyvinylidene fluoride (PVDF), polyterafluroethylene (PTFE) or styrene-butadiene rubber (SBR) (the latter being used in conjunction with carboxymethyl cellulose (CMC))

The fluorinated polymers, in particular, potentially pose environmental risks at the end of the cell's life [183,189]. CMC is a great alternative that, as of right now, is mostly used in the production of anodes for lithium ion batteries but has also found use in other systems [190–193].

Binder free electrodes have also occasionally appeared in the literature as an effort to produce a greener battery [194,195]. Some of these systems, however, would be uneconomical in the long run.

6.3.4. Coating

Currently, the industry standard for the coating of the slurry onto the current collector is by dispersing the dry mix of active material, binder, and conductive carbon into a solvent, commonly N-Methyl-2-Pyrrolidone (NMP) [196]. This is a hazardous substance, and it requires energy intensive drying, solvent recovery, and purification systems, all of which require significant capital expenditure. Moving away from toxic solvents toward benign solvents, or even dry electrode coating techniques can help reduce the cost and the environmental impact of the production of batteries.

Wood et al. compared water to NMP and found that using either solvent did not significantly change the contribution of cost to a battery pack (both being roughly 8–9%). However, using water as a solvent did reduce the post coating processing by saving two million dollars in capital expenditure [196].

Another popular technique that has started picking up the traction of late is the dry electrode coating. The dry coating comes with a myriad of benefits. It avoids the energy intensive drying process entirely, thus lowering capital cost requirements, is environmentally benign, and allows for comparable or even enhanced performance compared to traditional coating techniques [197–200].

7. Electrolytes

Electrolytes are a key research area that cannot be overstated in the development of effective batteries. An ideal battery electrolyte must have the following attributes;

- *Electrochemical window:* Electrolytes need to be stable over a wide voltage range that cell reactions can be accommodated without electrolyte decomposition.
- *Conductivity:* High ionic conductivity and low electrical conductivity is desired. A few mS/cm is the minimal ionic conductivity considerable for battery electrolytes.
- *Viscosity:* Electrolytes with low viscosity have higher ion mobility.

- *Thermal stability:* Thermally stable electrolytes, which are liquid over a wide range of temperatures, are ideal. Solvents with high flash points are safer.
- *Wettability:* The electrolyte should also wet the electrodes and the separator well so as to achieve consistent ion mobility.
- *Cost:* Low cost and environmentally friendly electrolytes should be preferred.
- *Compatibility:* Noncorrosive, nonreactive electrolytes ensure the longevity of cell components.

Moreover, electrolyte design and composition should be carefully coordinated with the nature of the electrode surface. Altering the electrolyte can be one of the simplest ways to resolve problems associated with high energy electrodes. For example, the addition of fluoroethylene carbonate has been shown to limit dendritic growth with the formation of a stable LiF solid electrolyte interphase (SEI) layer on the Li anode [21]. Goodenough et al., have shown the addition of bis(4-nitrophenyl) carbonate additive can precipitate out the troublesome polysulfide species in Li−S cells, almost entirely eliminating the shuttling phenomena and allowing Li−S cells to retain over 700 mAh/g after 300 cycles [35]. Adding an extra chemical into an existing cheap, high-performance electrolyte is a low-cost mehtod to improve performance so long as the additives are not expensive or are only a small proportion of the overall electrolyte. Common electrolytes can be classified as organic, aqueous, water in salt, ionic liquid, or solid.

7.1. Organic electrolytes

Organic electrolytes are the most well-known form of electrolytes. These are what enable the batteries in our cellphones, laptops and increasingly, vehicles. A typical organic electrolyte involves dissolving a metal salt, which employs a large, and hence, weakly coordinating anion, such as BF_4^- or PF_6^- to a standard 1 mol/dm^3 concentration. Typical ionic conductivities in organic solvents are around the 10 mS/cm mark [201]. Preference for organic solvents in electrolytes can be attributed to their wide electrochemical window compared to water.

Historically, organic carbonates, such as ethylene carbonate (EC), propylene carbonate (PC), dimethyl carbonate (DMC), and their mixtures have been used for Li-ion batteries. These solvents, especially EC, are mandatory for 4 V operation of Li-ion batteries [202]. The passivation layer formed by EC and its high flash point, make it an essential component in carbonate solvent mixtures. However, carbonates cannot be utilized for high voltage ($>$ 5 V) systems, as this is beyond the limit of their anodic stability [203]. Various esters, ethers, and sulfones have also been used in Li batteries [204−206]. Flamme et al. have reported a comprehensive study on the selection of organic solvents for Li-ion batteries, in which they assess different solvents based on their physicochemical properties, as well as environmental impact and production pathways. They conclude that the introduction of nitrile and fluorine groups can increase stability while alkoxy group insertions help reduce the melting point and increase conductivity. Also, general desymmetrization of solvent species increases solvation and decreases melting points [203]. The same classes of organic solvents have also been studied for other monovalent ion batteries, such as Na^+, K^+ ion batteries [207,208].

The biggest downside to these materials is their flammability. Safety is a major consideration when commercializing a battery technology, and as advanced ion batteries become more and more ubiquitous, finding an electrolyte that is not inherently dangerous is of utmost importance. Noteworthy here are the solvents with high flashpoints, such as Gamma-butyrolactone (95.5°C) and sulfolane (165°C), which can be safer alternatives to carbonates [203]. Such safer solvents that can be produced via cost-effective pathways and that which leave a small environmental footprint are needed.

7.2. Aqueous electrolytes

Aqueous electrolytes have the potential to be far cheaper than their organic counterparts. However, there is a large trade-off when considering the voltages that are able to be achieved. Theoretically, at potentials above 1.23 V water begins to “split” into hydrogen and oxygen gases. In practice, this electrochemical window can be extended by using electrode materials that sufficiently deactivate the oxidation and reduction of water. For example, the lead-acid battery has an aqueous electrolyte that is able to operate up to 2 V since water electrolysis on the lead oxide electrodes is sufficiently slow and can be slowed further by the introduction of additives, such as benzaldehyde [209,210].

In general, however, staying within the electrochemical window of water severely limits to operating voltage of cells to around 1 V. For a system with identical capacity, this represents a four-fold decrease in the energy density compared to traditional ∼4 V Li based systems.

In spite of this, there is still a sizeable literature presence of aqueous batteries [211−216]. One of the major challenges for this field is finding electrode materials compatible with water and whose redox reactions occur within the limited electrochemical window. For example, the intercalation of lithium into graphite occurs at ∼ -2.9 V vs. SHE. This is well below the thermodynamic potential of hydrogen evolution (0.00 V vs. SHE), and so graphite cannot be used as electrode material in aqueous lithium-ion batteries.

7.3. Water in salt electrolytes

Although organic electrolytes can provide excellent energy densities due to their wide electrochemical window, there are safety concerns regarding their composition being made of flammable and toxic materials, which can limit their application [217]. Aqueous electrolytes solve the flammability issue, but the electrochemical window is severely reduced, limiting energy density and commercial viability. Water in salt electrolytes benefits from wider electrochemical windows than traditional aqueous electrolytes. While aqueous electrolytes are limited by the splitting of water to 1.23 V, Suo et al. showed water in salt electrolytes can have electrochemical windows as wide as 3 V [218]. This gives them the ability to facilitate the typical operation voltages of Li anode materials and produce high energy cells. Water in salt electrolytes is defined by the mass and volume of the salt far exceeding that of water. Suo et al.'s electrolyte was comprised of LiTFSI dissolved at extremely high concentrations (molality $>$20 m) in water. This results in all water molecules being a part of ion solvation shells, with no free water present. This is what allows for higher operating voltages compared to traditional aqueous electrolytes, which have copious amounts of free water present, and is, thus, limited to water's electrochemical window. However, this water in salt electrolytes is inherently expensive. Li is a rare element, comprising roughly 18 atoms in every million of the Earth's crust [16]. Most organic electrolytes contain about 1 M of Li salt, so upscaling to tens of times more Li is not a cost-effective method going forward. Not to mention the increased toxicity due to the increased concentration of Li salts. LiTFSI is toxic upon contact or ingestion and can cause damage to organs. It is also corrosive to the skin and eyes [219]. One way Lukatskaya et al. circumvented these downfalls was to use the very water-soluble potassium acetate (KAc) in a mix with lithium acetate (LiAc). These are both benign and reduced the cost

of the electrolyte by an order of magnitude, all the while providing the same extended voltage window [220]. Conductivity was also excellent, being as high as 31.4 mS/cm in the case of 27 m KAc, without LiAc. This is a much higher conductivity than traditional organic electrolytes, which can achieve roughly 10 mS/cm at room temperature [201]. Water in salt electrolytes, therefore, shows great promise in providing nonflammable, cost-competitive cells, particularly for applications where safety is paramount, such as in aviation.

7.4. Ionic liquid electrolytes

Ionic liquids (ILs) have a variety of both pros and cons. Due to being a pure ionic mixture, they have a higher conductivity than common organic electrolytes due to the sheer abundance of ions, similar to water in salt electrolytes. For example, a typical IL comprised of $AlCl_3$ combined with 1-ethyl-3-methylimidazolium chloride ($AlCl_3$/[EMIm]Cl) in a 1.3:1 ratio achieves 20 mS/cm at room temperature [101]. Not quite as impressive as water in salt electrolytes, but still greater than their organic counterparts [201]. They also achieve a relatively large electrochemical stability window of about 2.45 V in the case of the typical $AlCl_3$/[EMIm]Cl electrolyte [52]. There are less widely used ILs, which can achieve much higher windows, such as 3.25 V in the case of aluminum trifluoromethanesulfonate (sometimes called aluminum triflate) with 1-butyl-3-methylimidazolium trifluoromethanesulfonate ($Al(OTf)_3$ /[*BMIm*]*OTf*), or even as high as 5 V when using piperidinium-based salts [221,222]. IL electrolytes are most widely used in Al-ion batteries. Due to the need to break through the passivating oxide layer, Lewis acidic electrolytes are needed.

In order to make highly conductive, runny IL electrolytes, awkwardly shaped cations, and anions are needed, where the overall charge is spread out over the molecule. These awkwardly shaped ions do not fit together easily, and thus, they have lower melting points than highly regular salts with a comparable size between ions, such as NaCl (m.p. 801 °C). Currently, in the Al-ion literature, $AlCl_3$/[EMIm]Cl is the standard, representing about 2/3s of all papers [18]. However, [EMIm]Cl, along with the other higher voltage alternatives are expensive, which severely limits commercial potential. As of September 2019, Sigma Aldrich stocks [EMIm]Cl at 98% purity for $87.30 USD/25g. Thankfully, there are ongoing investigations into much cheaper alternatives, such as imidazole hydrochloride, which Sigma stocks at $125.00 USD/500g. Xu et al. achieved an impressive 105 mAh/g after 1000 cycles at a high current density of 4 A/g using imidazole hydrochloride coupled with $AlCl_3$ [223]. This is very impressive for a $AlCl_3$-graphite cell, where typically graphite achieves sub 100 mAh/g capacity [52,224]. More research into cheaper, high voltage window electrolytes is needed if cheap, economically viable ILs are going to be commercialized for use in battery systems. Particularly, in the case where active ions have a highly negative reduction potential.

7.5. Solid electrolytes

At first glance, the concept of a solid electrolyte seems strange. If the main role of an electrolyte is to facilitate the flow of ions between the electrodes, then surely liquids would be the natural choice. However, liquid electrolytes do have a few flaws. They can be unstable, both electrochemically and thermally, and they can be expensive and unsafe in many cases. The use of solid electrolytes can also aide in the use of high capacity materials, such as metallic anodes and sulfur cathodes, by suppressing dendrite formation in the case of the anodes, and polysulfide shuttling in the case of sulfur [225].

The requirements for a good solid electrolyte are much the same as for traditional liquid electrolytes; low cost, high safety, and compatibility with electrode materials are all essential. However, with solids, one must also consider the conductivity of the material. Specifically, the solid electrolyte must be able to quickly conduct ions, but at the same time not be able to conduct electrons. While these properties are somewhat inherent to liquid electrolytes, finding a solid that fits both of these properties can be challenging.

While many different materials fitting some of these categories have been discovered. A full discussion of the field is outside of the scope of this article, and so only the cheapest and abundant materials will be covered.

If a material that has all of these properties is found then there are significant advantages for these systems compared to traditional liquid-based electrolytes. Most importantly, they often allow the use of metal anodes, which is highly desirable (see section 6.1). Not only will the electrolyte not react with the metal and passivate the surface, as is common with liquid electrolytes, it will also suppress the formation of dendrites, which is one of the main reasons that metallic anodes are difficult to work with [22,74]. However, recently Li dendrites have been observed in a garnet-type solid electrolyte by Ren et al. [226]. They found that the dendrites were able to grow through to interconnected pores and short the cell. They suggest that these inorganic materials be kept as free of pores as possible to prevent this from occurring. Solid electrolytes fall into two main categories, inorganic and organic.

7.5.1. Inorganic solid electrolytes

Inorganic solid electrolytes were the first to be investigated from both a fundamental and energy storage perspective. This is because the mechanism of ion conduction through inorganic solids is easier to understand and investigate. The possible mechanism for ion conduction has been explored in great detail. They range from the vacancy mechanism, whereby the ions hop from one vacancy in the lattice to the next, or the interstitial mechanism, where the ions hop from one interstitial site to the next [227].

The most famous and successful inorganic solid electrolyte is β-alumina, a polymorph of Al_2O_3 with an unusual layered structure. In fact, a battery using this material was commercialized as early as the 1960s. The high-temperature Na—S battery uses β-alumina as an electrolyte. A liquid electrolyte would not work in this particular instance because, due to the high operation temperature, both anode and cathode material is molten [83]. This increased temperature also helps to increase the rate of diffusion of the ions through the solid, which is advantageous.

This system also reveals another major advantage of solid electrolytes moving forward. Making a functioning metal-sulfur battery is one of the main goals of the battery community at large. However, there are several roadblocks hindering the development (see Section 6.2.1), in particular, the shuttling of polysulfides resulting in irreversible capacity loss, as well as other problems [35,225]. Polysulfides are generally too large to diffuse easily through the lattice, and so this loss pathway is shut down.

Another important class of inorganic solid electrolytes is the Sodium Super Ion Conducting or NASICON type materials. These generally take the form of $AB_2(PO_4)_3$ formula where A is a monovalent cation, Li, Na, or K and B is a metal ion generally Ti or Zr [228].

The main issue with inorganic electrolytes is the contact with the electrode [229,230]. Maintaining good contact is crucial for fast charging and discharging, and if this contact is disrupted, then the cell is dead. This can be worsened further in situations where the electrodes undergo large volume changes. The constant swelling and shrinking of the material can break this contact [229,230].

Because these materials are inherently not very flexible, there is not much that can be done to improve this.

7.5.2. Organic solid electrolytes

The use of an organic material as an ion transport medium was first demonstrated in 1973 by Fenton et al. [231]. It was shown that poly(ethylene oxide) (PEO) was capable of chelating the ions, which can then hop from site to site and move along the chain [232].

Since then, many different polymers have been studied for this purpose. However, as usual, most of the research is done in respect of Li-ion systems. Poly(acrylonitrile), poly(vinylidene fluoride) and poly(methyl methacrylate) have all been found as effective materials although PEO still remains the most popular [233,234]. Polymers have the advantage of being easy to work with and above all else, flexible. This feature opens up a wide range of possibilities for improved design architectures and implementation in wearable tech. This also solves the issues of inorganic solid electrolytes since they are more able to accommodate volume changes in the active materials.

Like the inorganic electrolytes, polymer electrolytes have to be used at an elevated temperature. Specifically, it must be above the glass transition temperature for the polymer [232]. However, polymers are much less thermally stable than inorganic materials, such as β-alumina. They are also much more easily oxidized compared to these materials, and so their operating voltage is generally lower as a result [235].

7.5.2.1. Summary of solid electrolytes. Solid electrolytes show potential and may come to challenge liquid electrolytes in the coming years. One of the biggest barriers to overcome is the elevated temperatures that are required for operation. This potentially rules them out of portable energy storage applications. Maintaining this high temperature must also be factored into the long-term energy efficiency of these cells.

Apart from the continued investigation of new materials, the biggest opportunities in solid electrolyte research are in composite systems. For example, constructing a cell with a polymer layer directly attached to the electrodes and an inorganic material sandwiched in between the polymer layers. This would theoretically improve the contacts between all components and make it more resistant to volume changes [236].

8. Conclusion

As we transition to a renewable energy economy, managing energy vectors by storing excess renewable energy becomes an imperative rather than an option. As a consequence, the demand for energy storage devices, batteries, in particular, will increase significantly. This increased demand will put a lot of pressure on battery commodities, which means that we can largely exclude exotic and rare materials as options for future low-cost battery technologies.

In this review article, we explored different battery materials, focusing on those that meet the criteria of future demand. Transition metals, such as manganese and iron, are safe, abundant choices for intercalation based cathodes, while sulfur has perhaps the highest potential for conversion cathodes.

Although alloys can provide significant capacity for the corresponding anodes, the associated volume changes require elaborate and often expensive solutions. Pure metal anodes offer still the highest potential performance. In order to use these anodes, problems, such as dendrite and/or solid electrolyte, interphase formation must be addressed. The simplest, and thus, potentially the most cost-effective method to tackle these issues is by exploring different electrolytes, either with entirely different types of electrolytes, such as solid electrolytes or with the addition of additives in order to suppress unwanted behavior.

Less glamorous cell components, such as binders, current collectors, and separators, should be continually assessed for economic viability and environmental friendliness.

Lithium-ion batteries represent the vast majority of the current market and research space; however, this boom cannot continue indefinitely due to the rarity of lithium (and cobalt). A trend in the research space toward lithium-free battery alternatives can already be observed. Future research should focus increasingly more on these alternatives as a renewable economy will require a battery market, which is much more diverse. For example, high-performing lithium-ion batteries will be used in areas where high energy density is required, whereas cheaper and more sustainable alternatives, such as sodium or aluminum-ion batteries, will be used in areas where cost and scale are more important.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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